

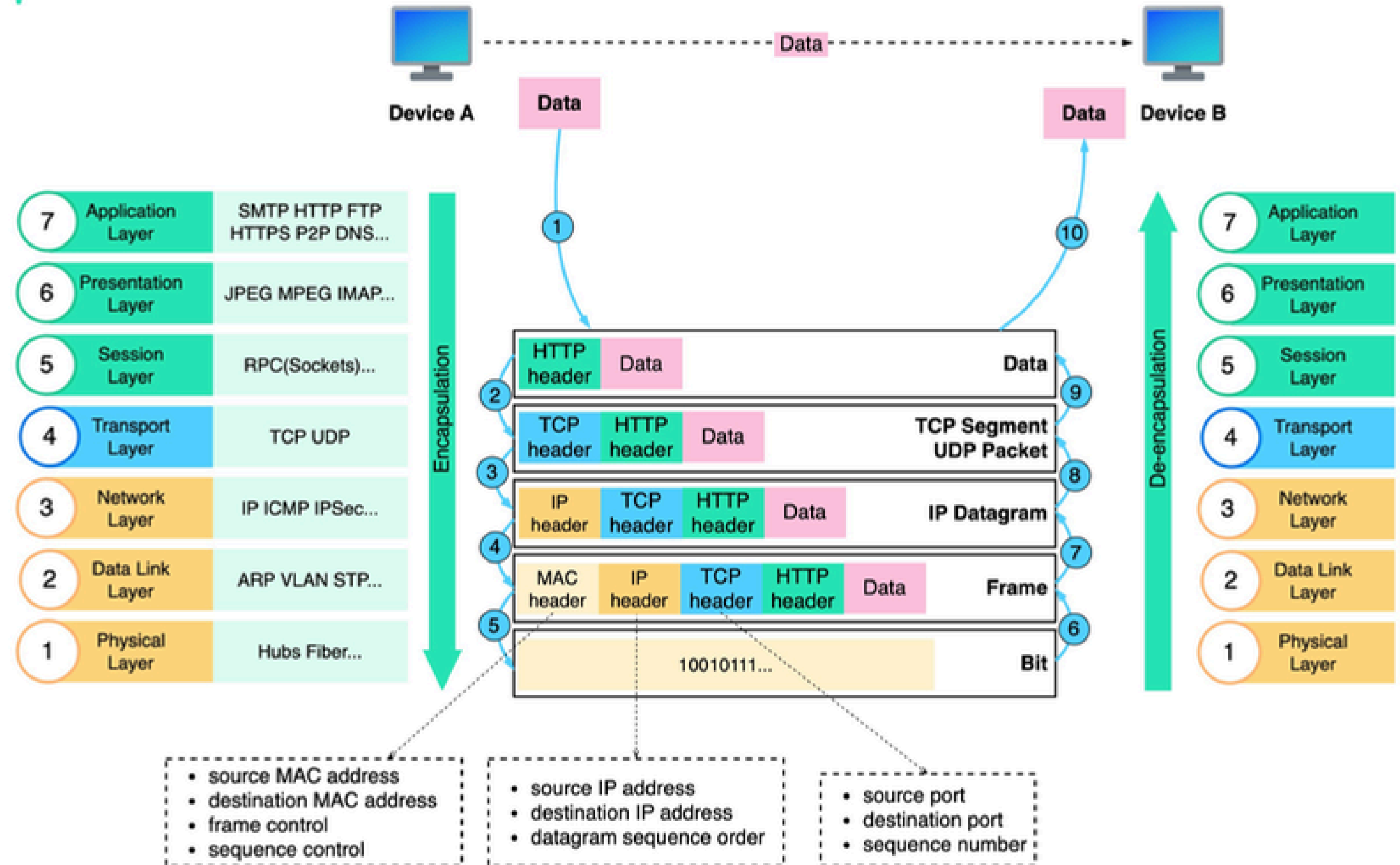


Email Protocols (SMTP, POP3, IMAP) CDNs and Caching

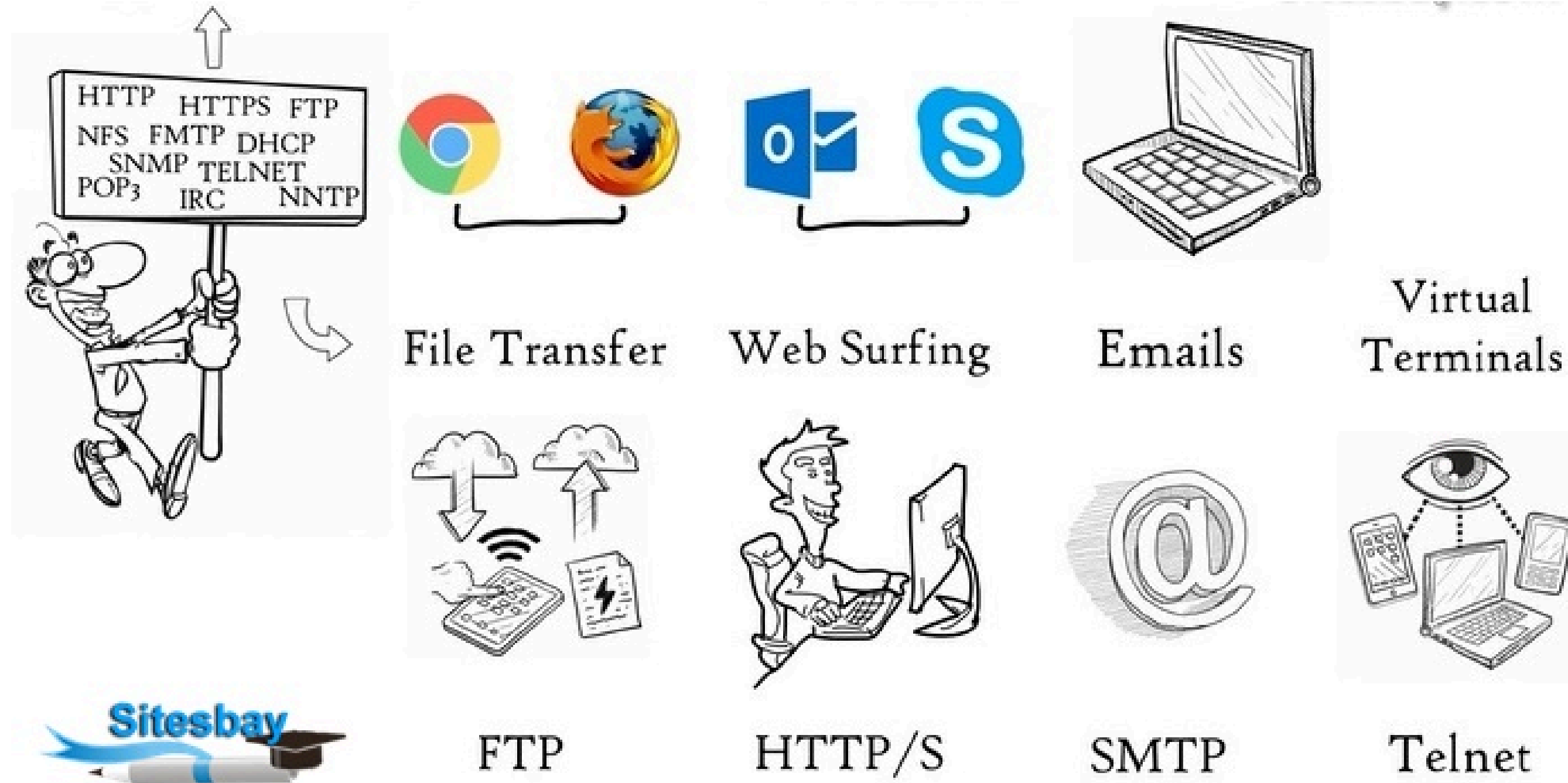
CSAI321:Computer Networks

Join the lecture online on your dashboard

Recap



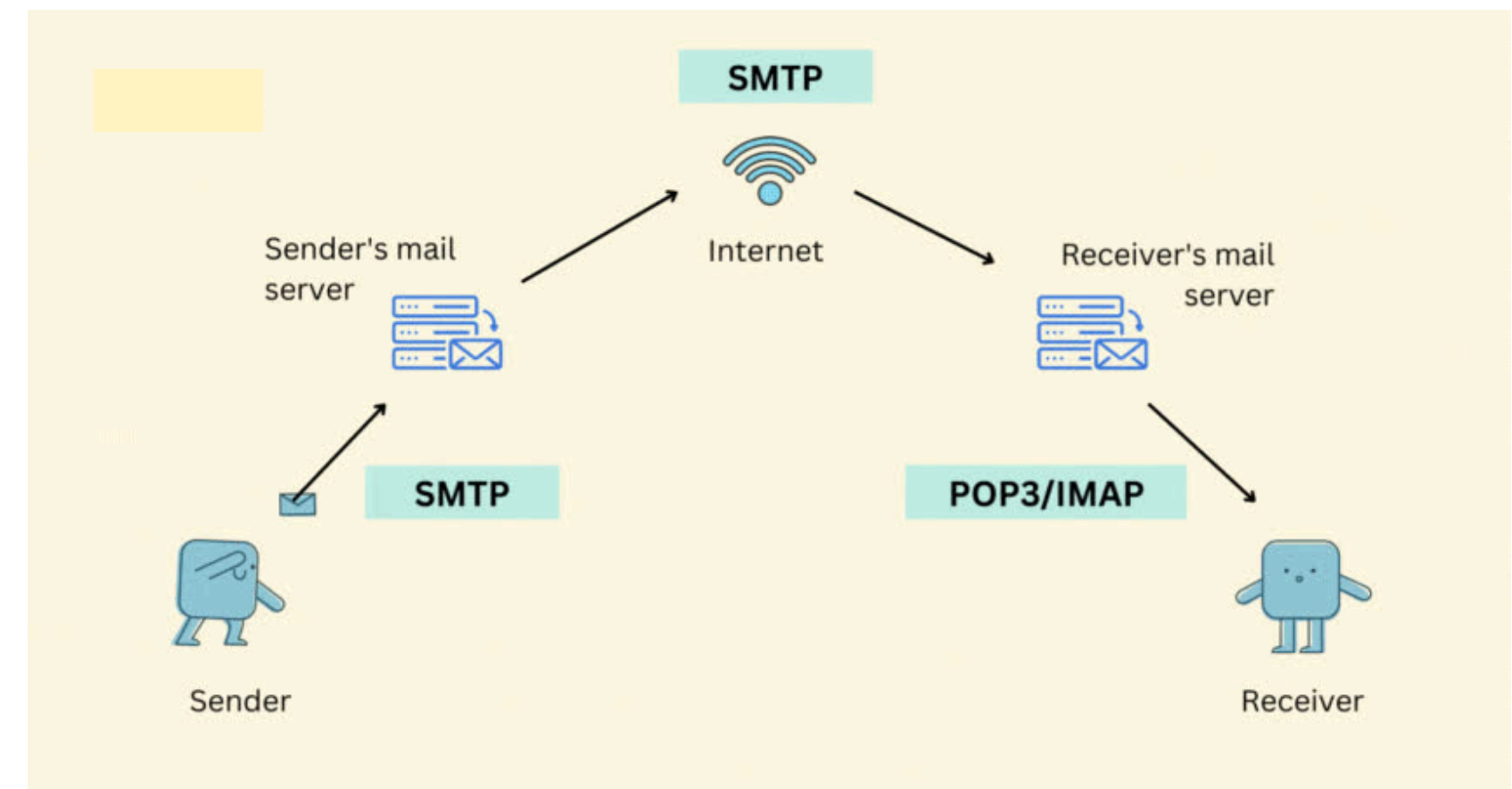
Application Layer Protocols



Email Server Protocol

How SMTP Works:

- The email client (e.g., Gmail app, Outlook) connects to the SMTP server.
- SMTP pushes the message from the sender to the receiver's mail server.
- The receiver then uses POP3 or IMAP to retrieve the email.

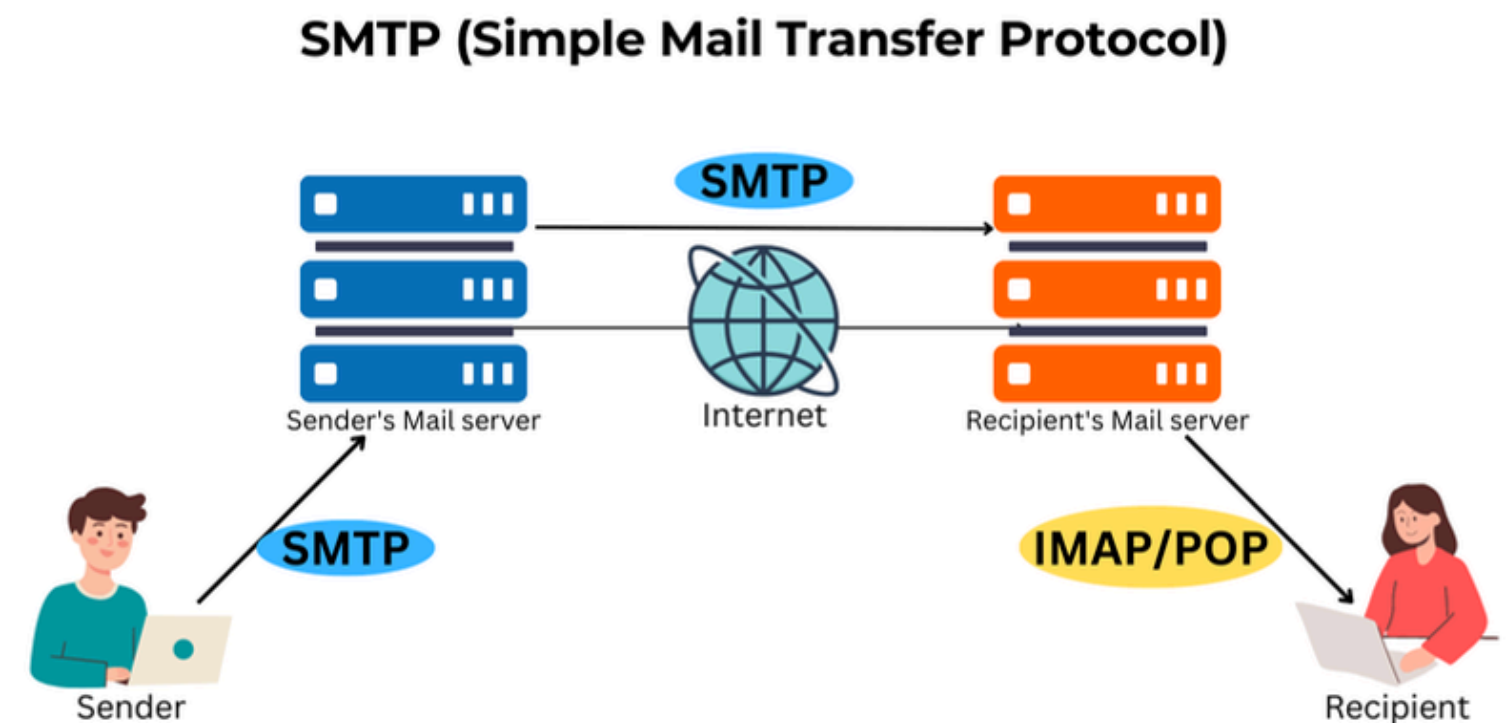


SMTP (Simple Mail Transfer Protocol)

- Sends email from sender server to recipient servers
- Push-based protocol
- Uses TCP, Ports: 25 / 587 / 465

SMTP in the Real World:

- Sendgrid, Mailchimp, Mailgun integrations use SMTP Relays



mailhog/**MailHog**

Web and API based SMTP testing



 37

 219

 16k

 1k



mailhog/MailHog: Web and API based SMTP testing

Web and API based SMTP testing. Contribute to mailhog/MailHog development by creating an account on GitHub.

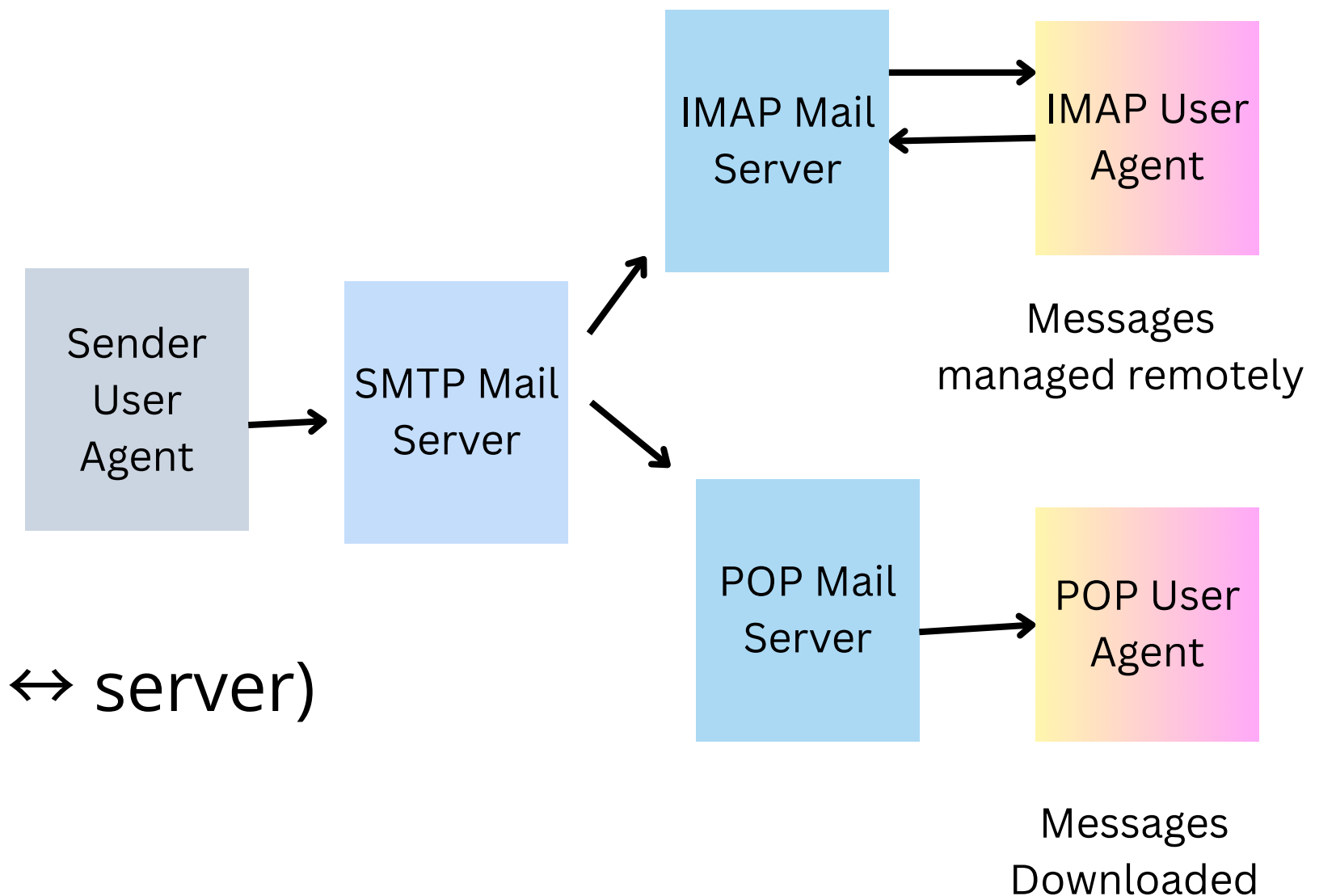
 GitHub

Mailhog

Email Clients Protocol

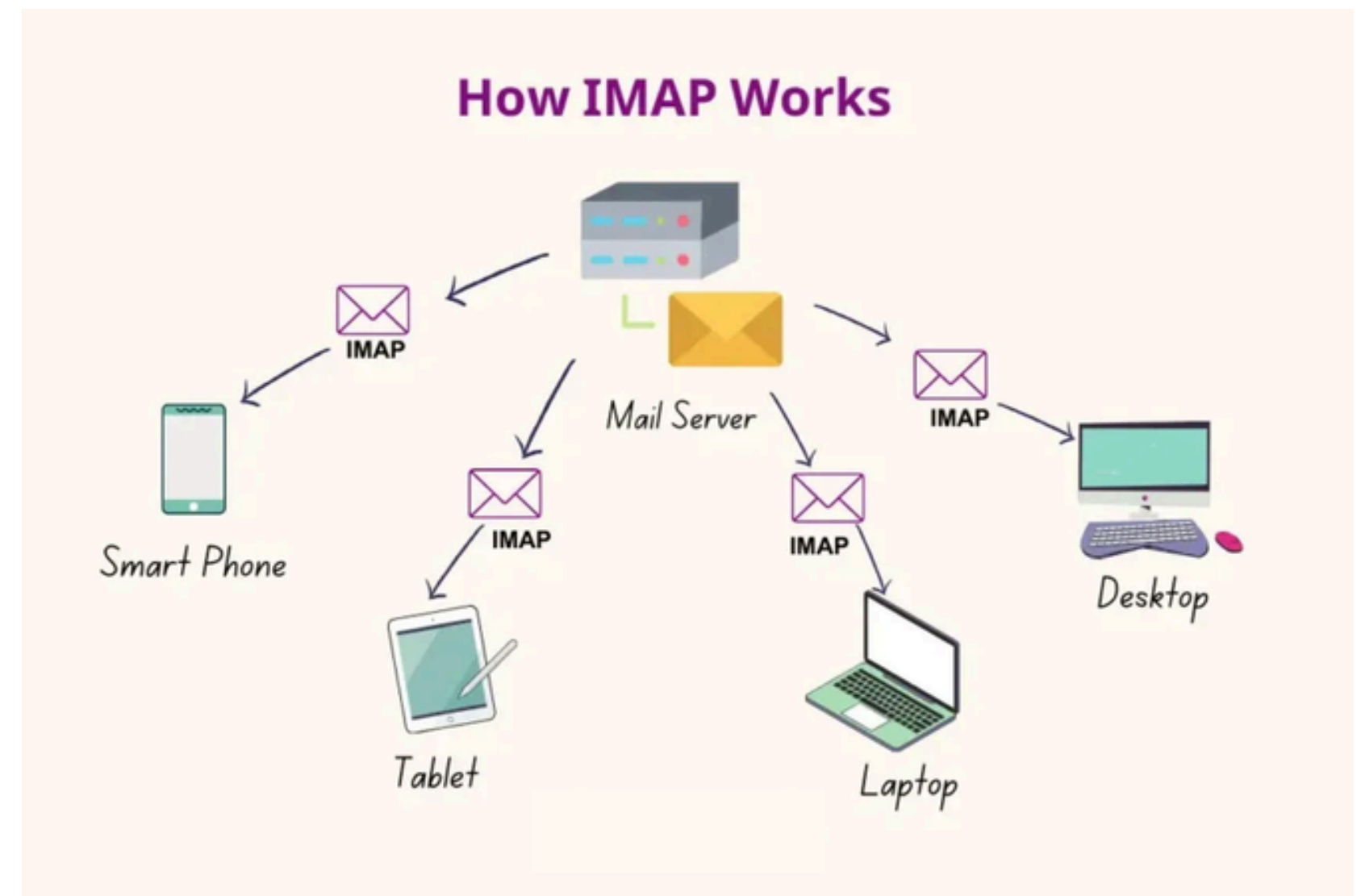
IMAP (Internet Message Access Protocol)

- Manages email directly on the server
- Suitable for multi-device access
- Supports folders, partial downloads
- Uses Port 143, Secure (IMAPS): 993
- **Connection Type:** Two-way sync (client ↔ server)



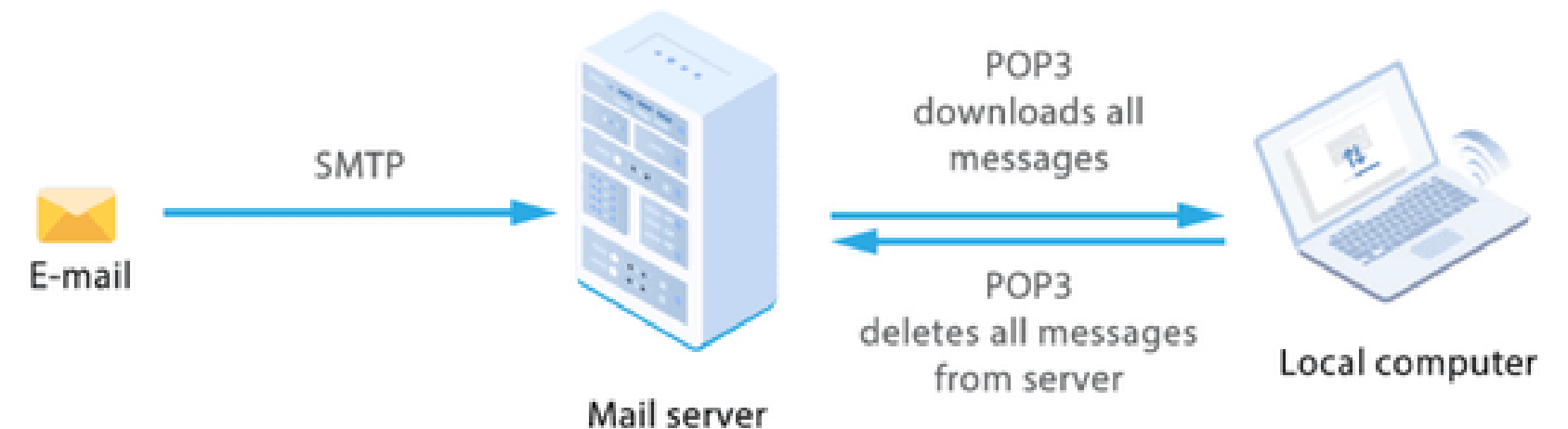
How IMAP Works:

- Your email client connects to the server using IMAP.
- You view and manage emails without downloading them completely.
- Changes (read/unread, folders, deletions) are reflected on the server and synced across all devices.
- You can access email folders (Inbox, Sent, Drafts, etc.) as organized on the server.



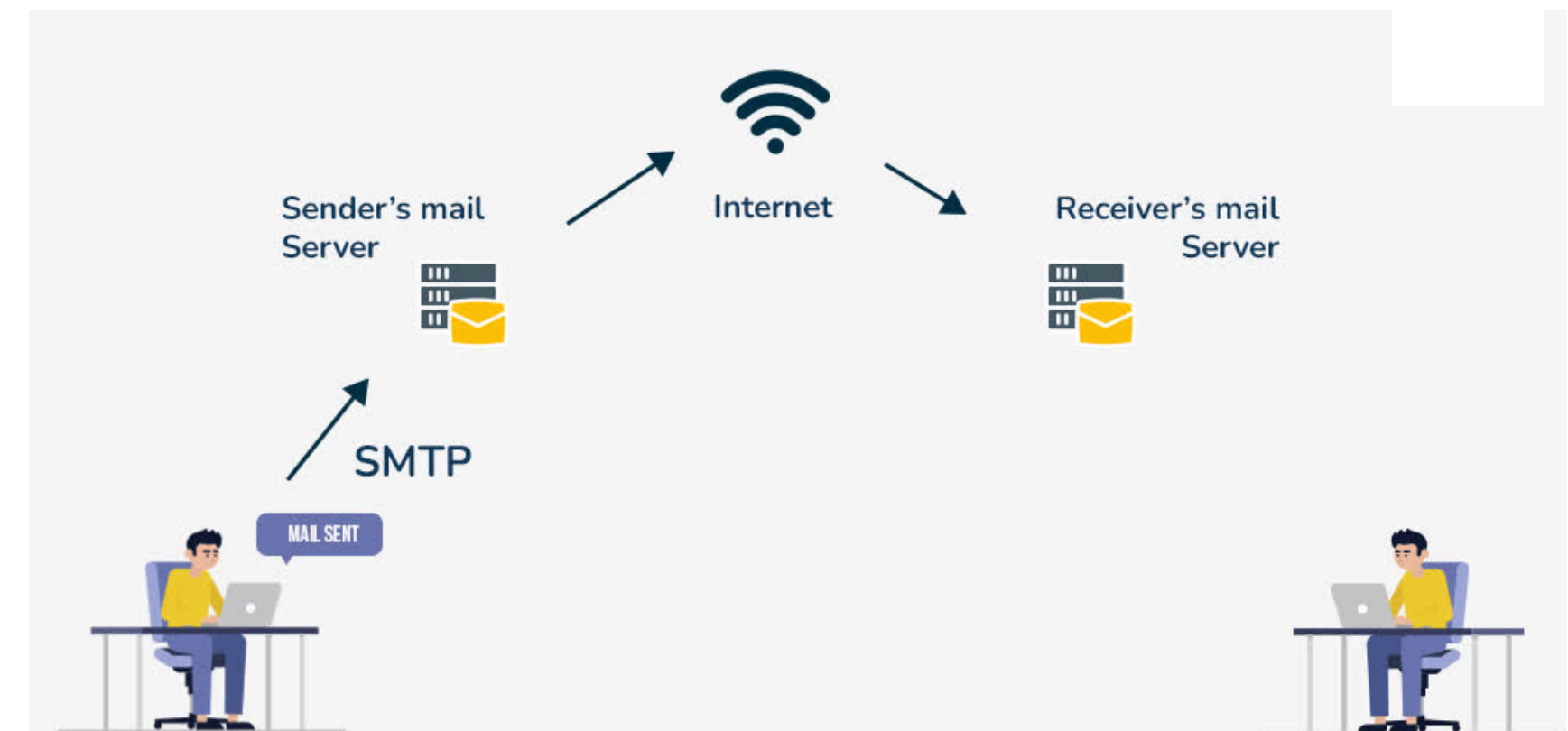
POP3 (Post Office Protocol v3)

- Downloads email to local client and deletes from server
- Best for offline access, but poor sync across devices
- Uses Port 110, Secure (POP3S): 995
- **Connection Type:** One-way (server → client)



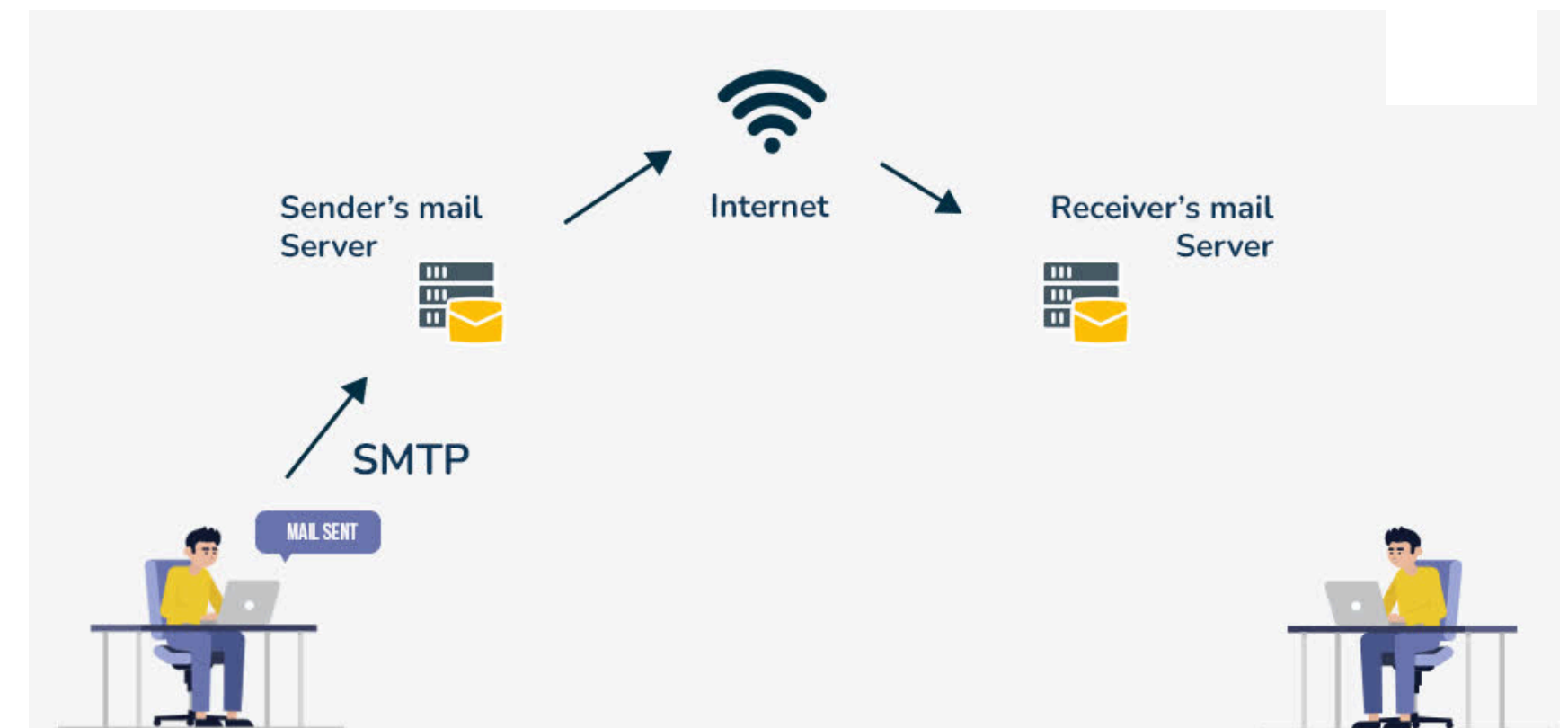
How POP3 Works:

- Your email client connects to the mail server using POP3.
- It downloads the emails from the server.
- It deletes the emails from the server after downloading.
- Emails are now available on your local device only.



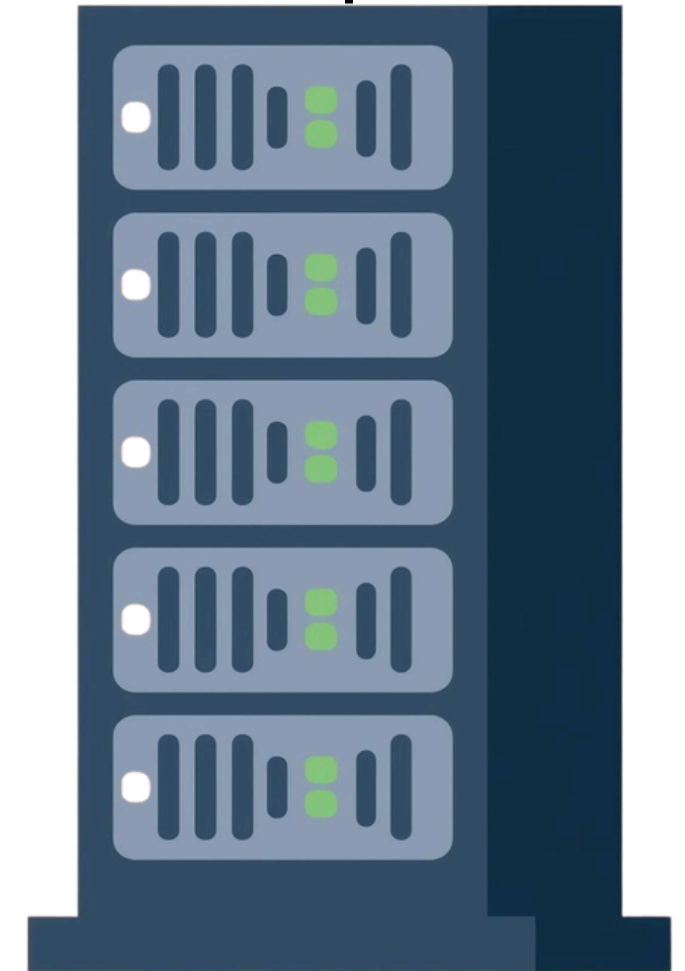
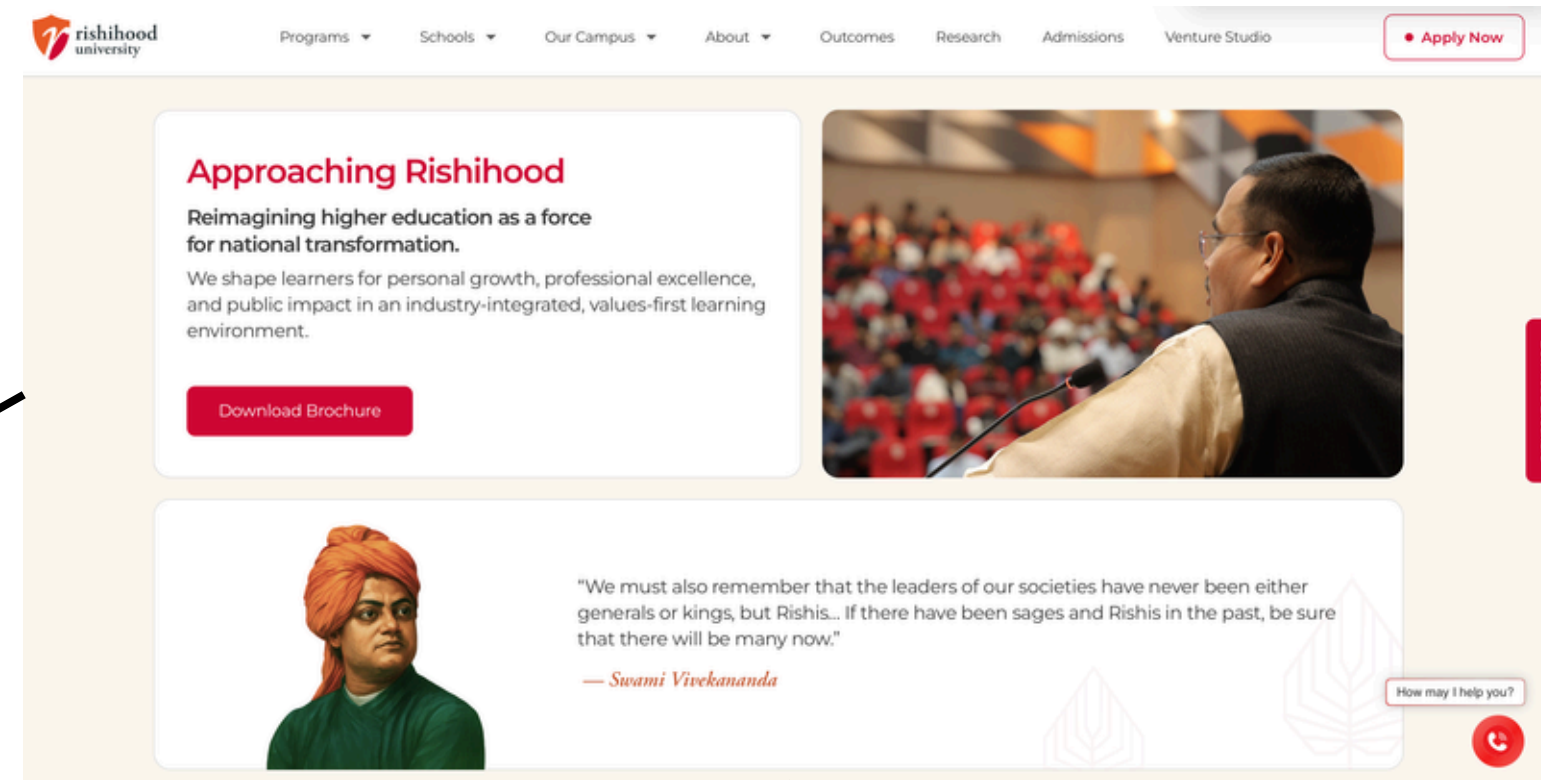
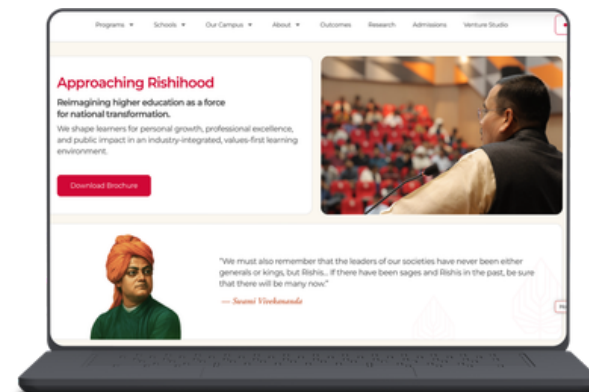
How POP3 Works:

- Your email client connects to the mail server using POP3.
- It downloads the emails from the server.
- It deletes the emails from the server after downloading.
- Emails are now available on your local device only.

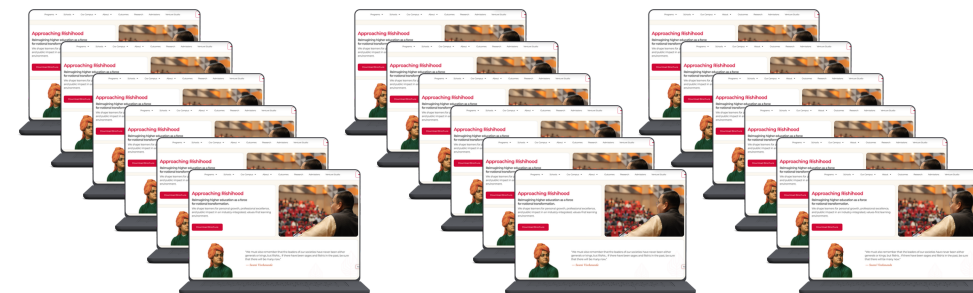
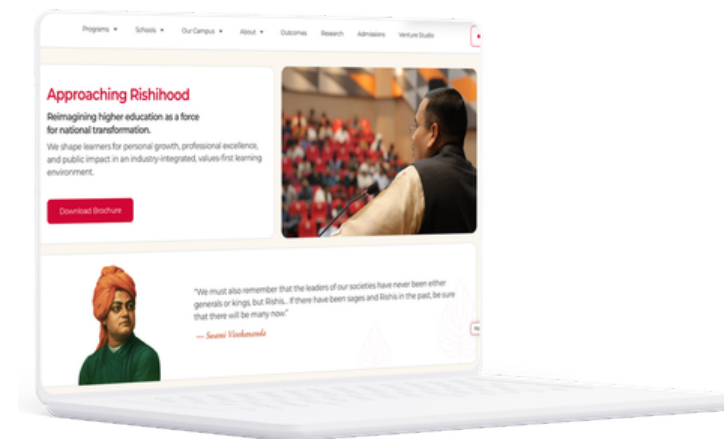


Application layer Quick and efficient Content delivering Methods

Imagine:

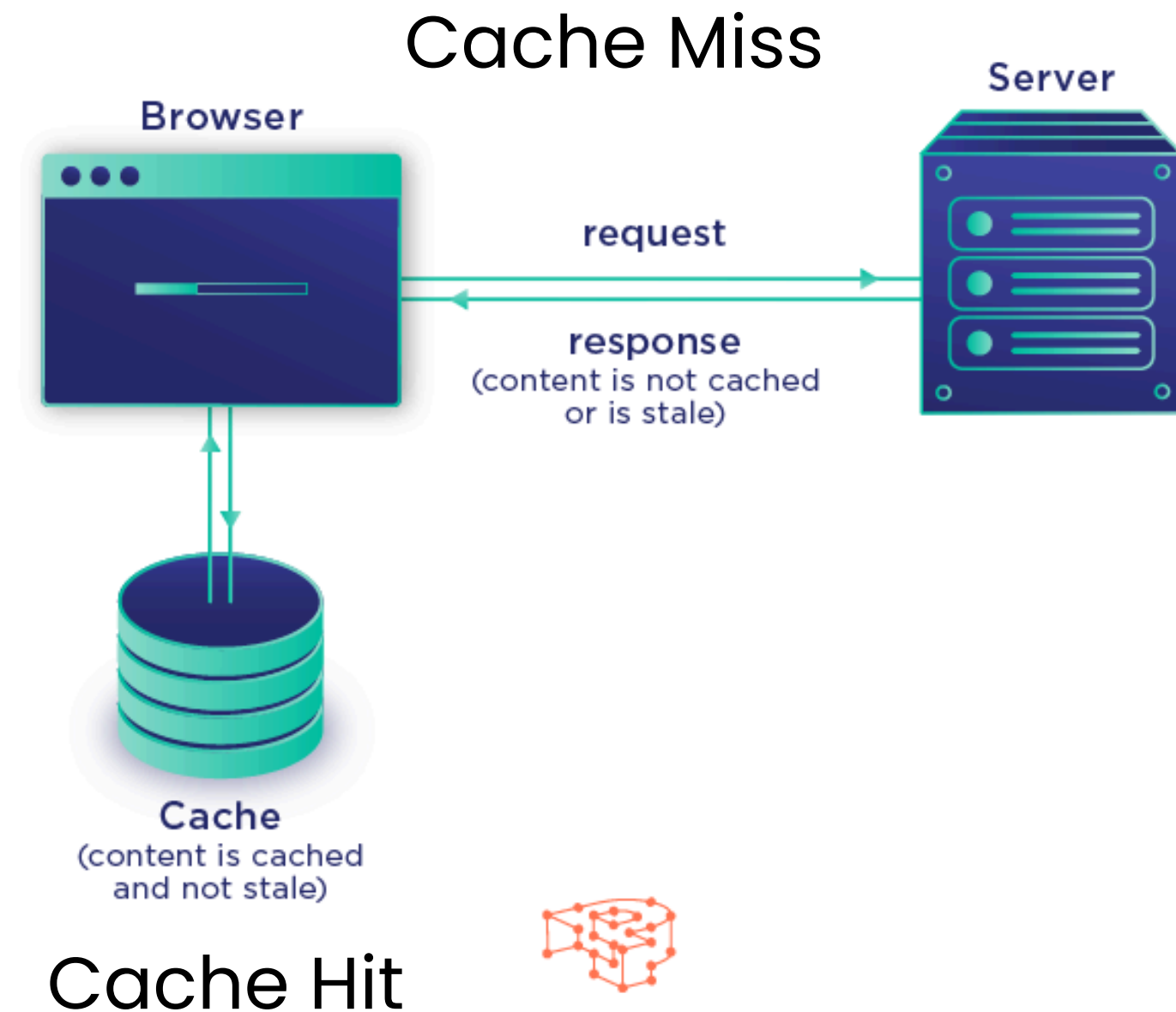


Same content
requested repeatedly.
Solution?



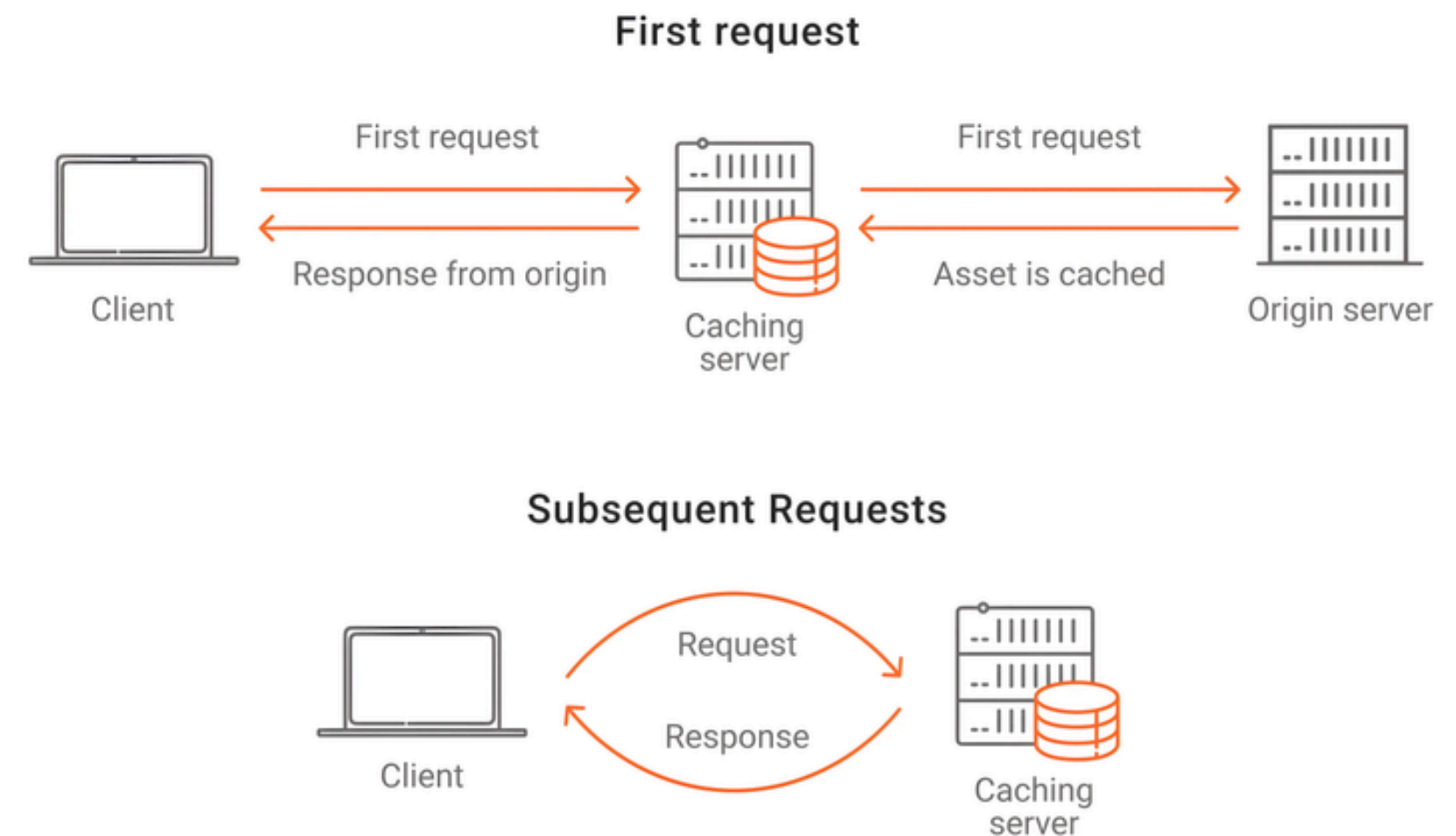
Caching

The process of **storing copies of data in a temporary, high-speed storage layer** so that future requests for that same data can be served much faster than fetching it from the original, slower source.



Caching Behavior

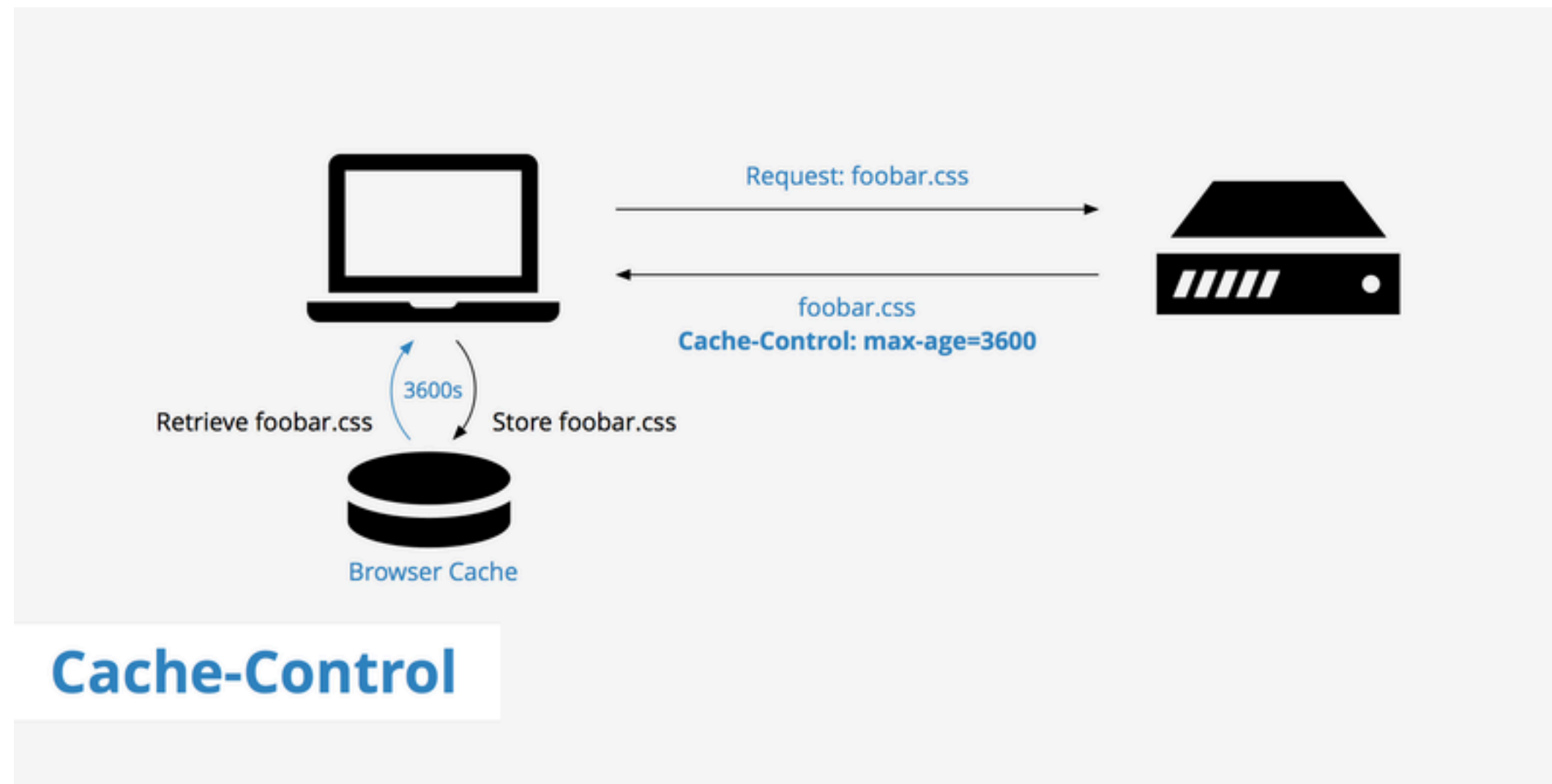
- Caching behavior refers to how data is stored temporarily (cached) and retrieved to improve performance and reduce latency.
- It's widely used in networking, browsers, web servers, CDNs, and applications.



Cache-Control

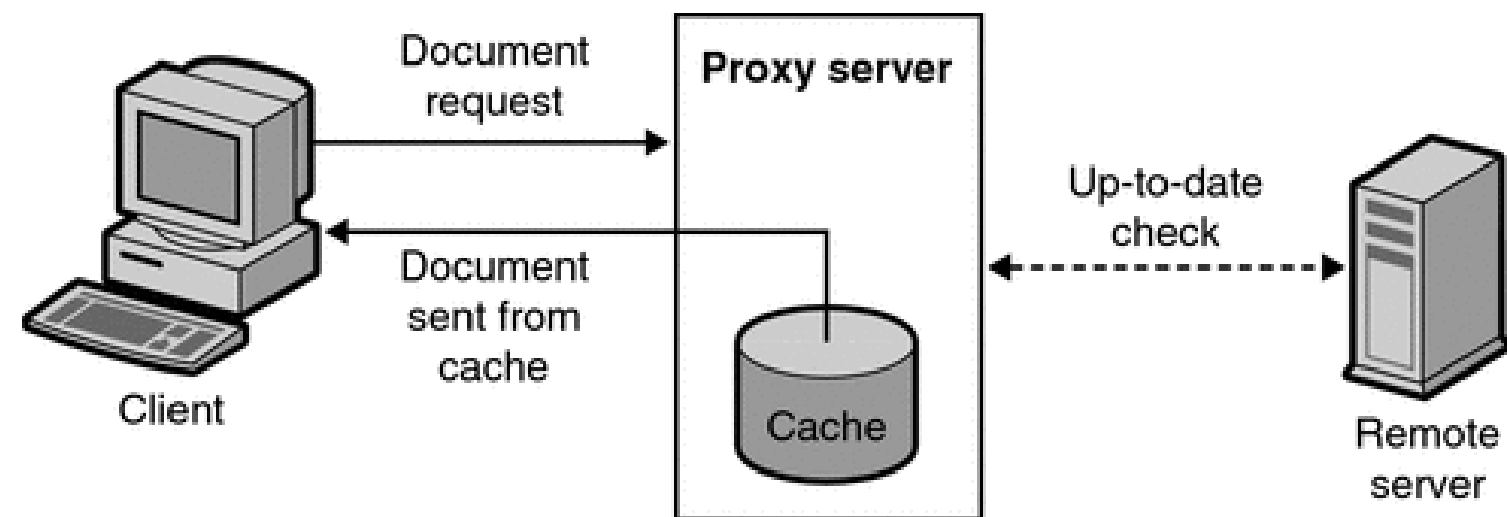
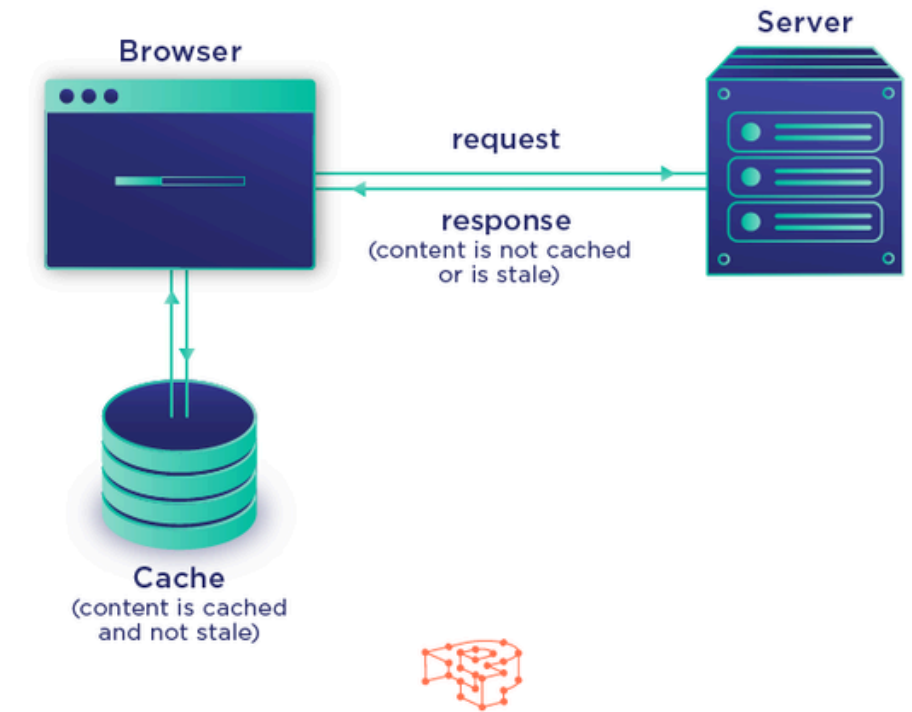
Format

Cache-Control: public, max-age=3600

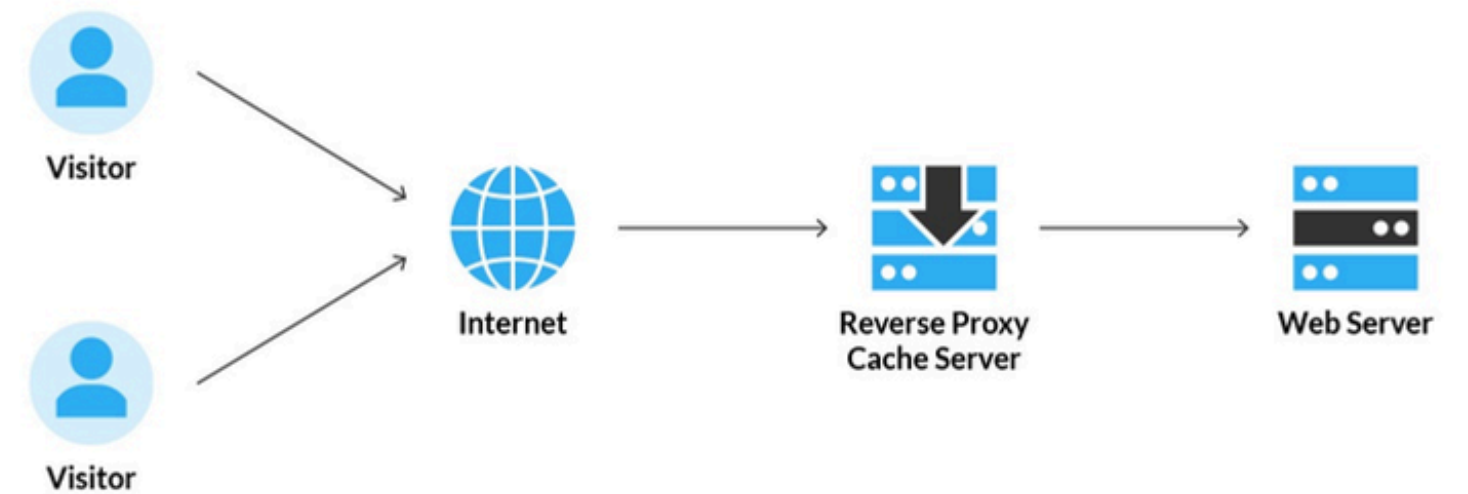


Layers of Caching

Browser Caching

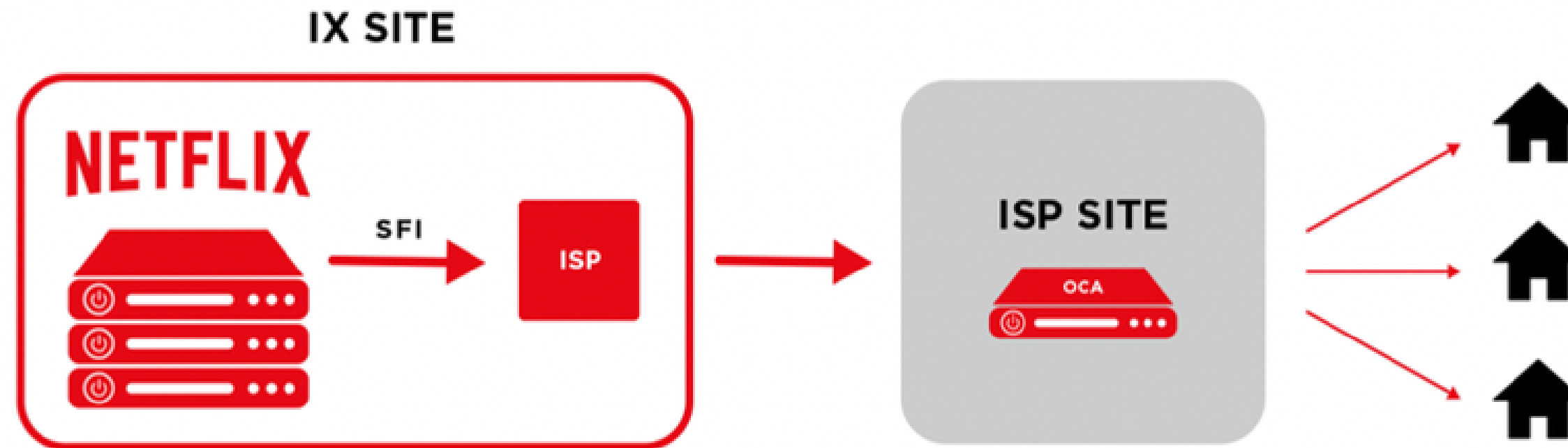


Proxy Cache



CDN Cache

Netflix Open Connect

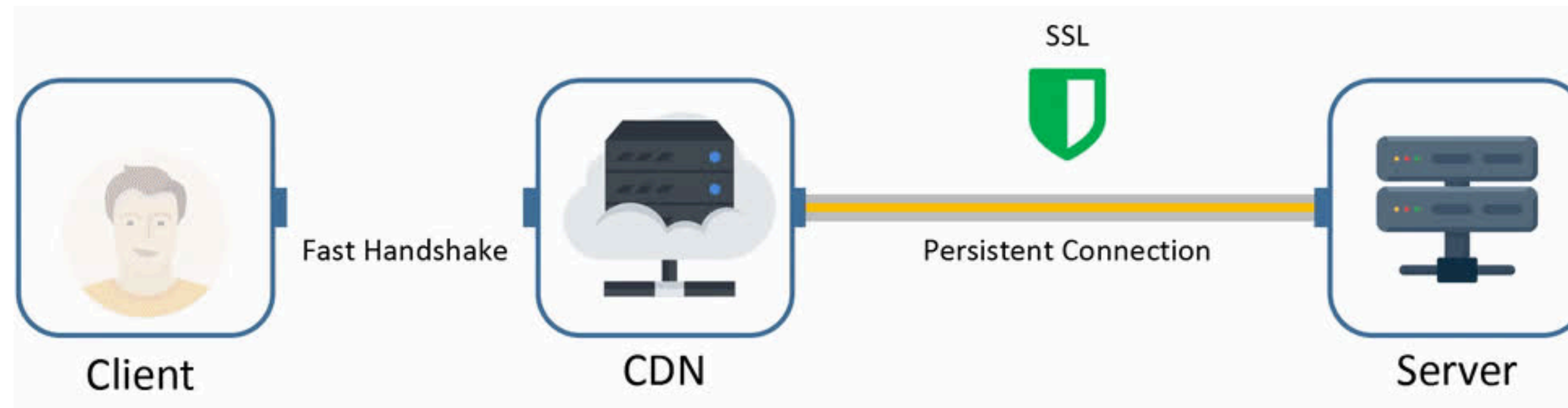


WOAH!

The storage in Netflix Open Connect Appliances at ISP sites (points of presence) holds up to 350 TB!

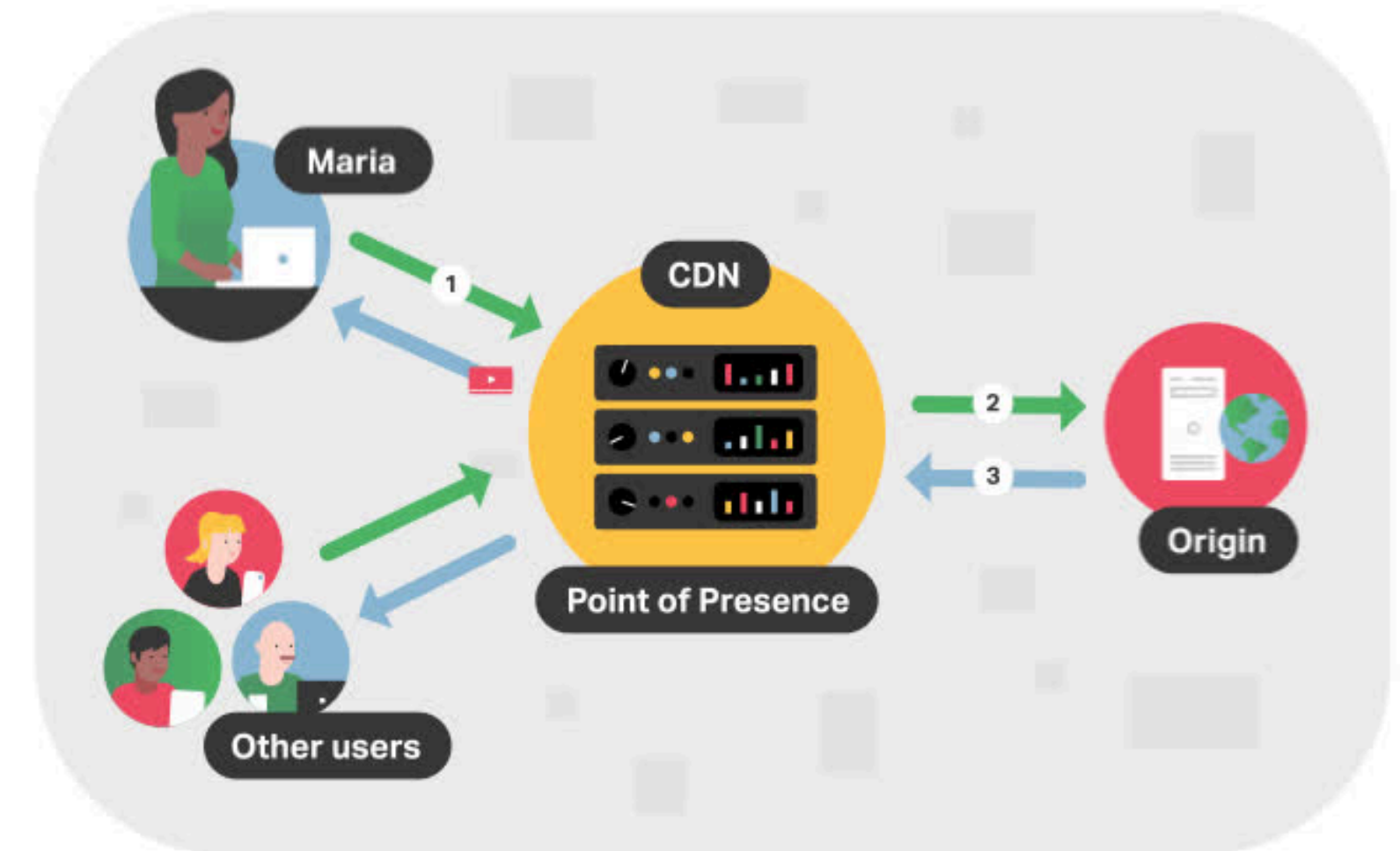
Content Delivery Network (CDN)

- A CDN is like a chain of grocery stores
- Instead of traveling to faraway farms, shoppers visit a local store
- Local stores stock food from distant farms, saving time and effort
- Similarly, CDNs cache web content closer to users
- Result: Webpages load faster, just like shopping takes minutes, not days

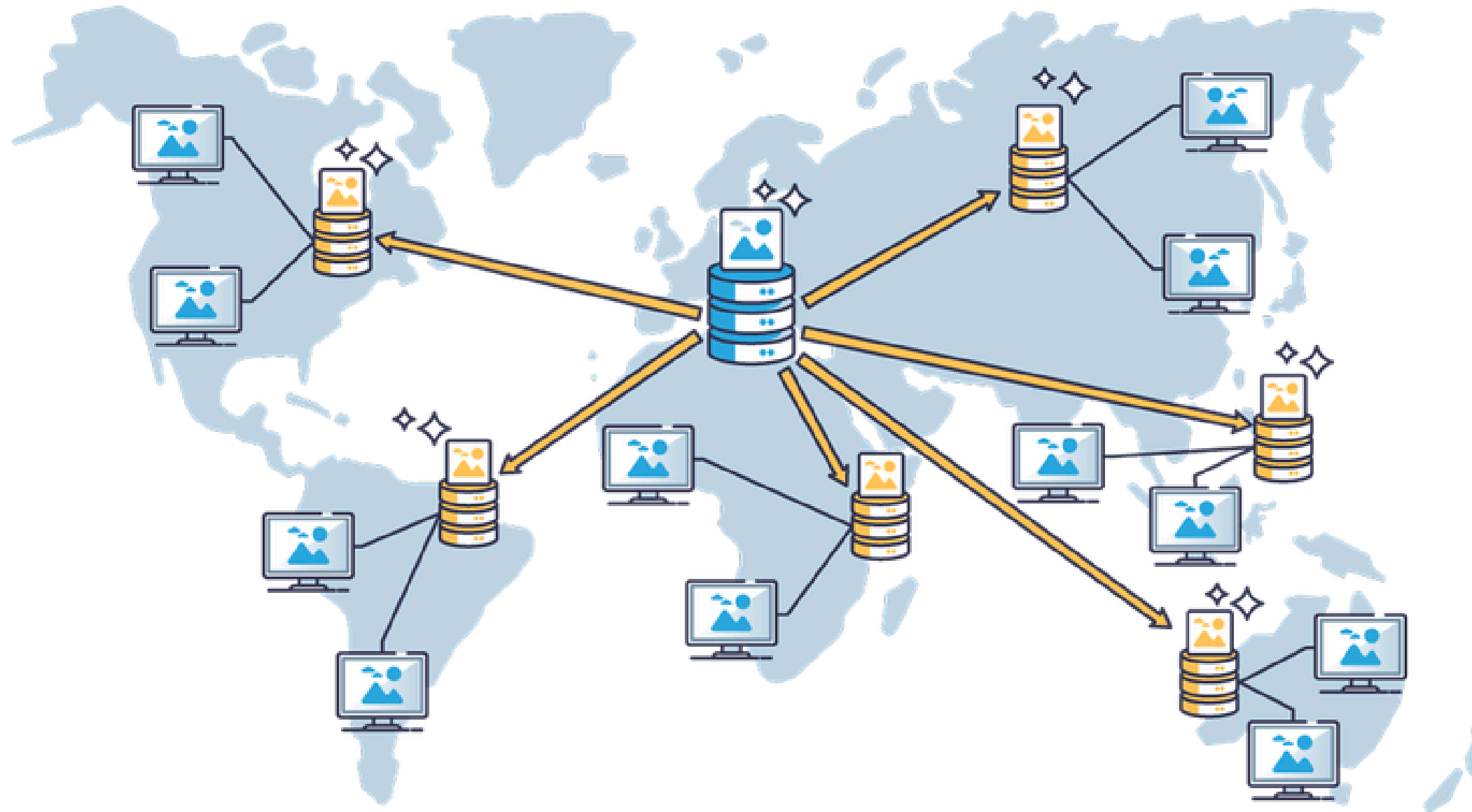


How a CDN Works:

- A user requests content (e.g., video or image) from a website.
- The nearest edge server serves the request instead of the origin server.
- If not cached, the edge server retrieves it from the origin and caches it.
- Benefits include faster load times, reduced bandwidth usage, and improved availability.



Content Delivery Network (CDN)



Geographically distributed servers, delivers content from the nearest edge server, reduces latency, saves bandwidth

Content Delivery Network (CDN)

Origin in the Late 1990s: CDNs emerged to solve web congestion and latency issues caused by growing internet usage and media-rich content, with Akamai pioneering the space in 1998.

Major Players Today: Key providers include Akamai, Cloudflare, Amazon CloudFront, Netflix Open Connect, Google Cloud CDN, Microsoft Azure CDN, and Fastly.

You can use a tool to verify what CDNs a particular website is using: [CDNPlanet](#)



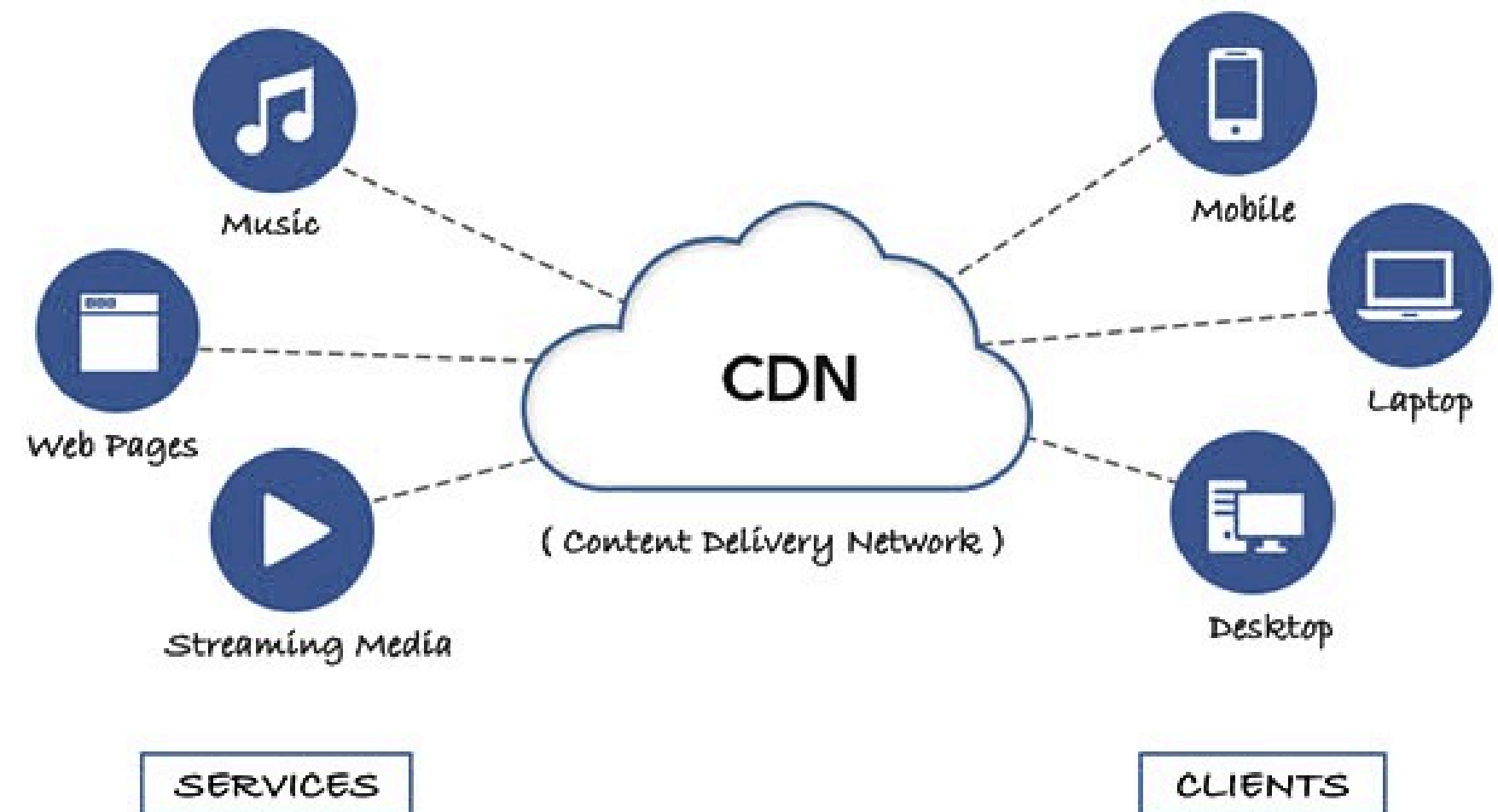
Content Delivery Network (CDN)

Use Cases

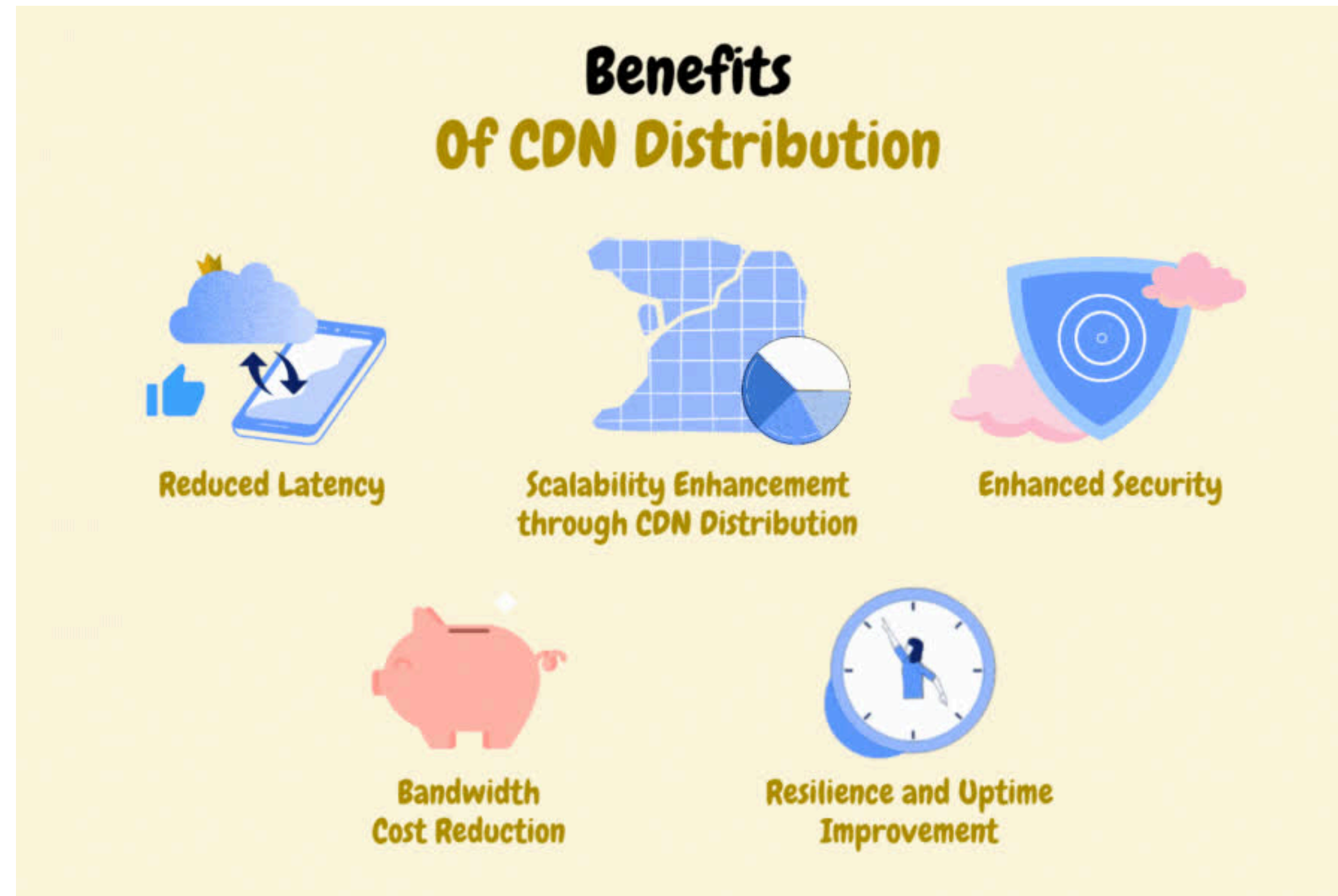
Websites, video streaming, software downloads, gaming, mobile apps

Examples

Cloudflare, Akamai, Amazon CloudFront, Google Cloud CDN



Benefits of CDN :



How Caching Works

- A client requests a resource (e.g., image, CSS file).
- The server responds and may include cache-control headers.
- The client stores the resource in its local cache.
- On the next request, the client checks:
 - Is the cached copy still valid (not expired)?
 - If valid -> use from cache (no network trip).
 - If expired -> revalidate or fetch again.

Fastly outage in 2021



- Fastly Outage Overview (June 2021): A misconfiguration in Fastly's network triggered a widespread outage, taking down major websites like Reddit, Amazon, and The New York Times.
- Impact on Internet Users: The outage highlighted how reliant the modern internet is on CDNs, as users experienced disruptions and slow load times across multiple platforms.

Please fill the feedback form.

**Thanks
for
watching!**